

CARA-FinSent Execution Guide

Objective

Generate experimental evidence for a publication-grade financial sentiment analysis paper. The target is not another accuracy-only benchmark. The target is a decision-grade system that adds retrieval, structured financial signals, agreement-aware training, calibration, abstention, and external validation.

Expected final evidence package

- classical baseline summary CSV
- structured feature ablation summary CSV
- retrieval experiment summary CSV
- agreement-aware training summary CSV
- calibration summary with ECE and Brier score
- CARA-lite integrated summary CSV
- prediction-level CSV files, confusion matrices, reliability bins, abstention curves, and PNG figures

Step 0 - Setup

Local setup:

```
python -m venv .venv
source .venv/bin/activate
pip install -U pip
pip install -r requirements.txt
python -m compileall -q src scripts
```

Docker setup:

```
docker build -t cara-finsent .
docker run -it --rm -v "$PWD:/app" cara-finsent bash
```

Colab setup:

```
!git clone <your-repo-url>
%cd cara-finsent-experiments
!pip install -r requirements.txt
```

Step 1 - Prepare datasets

Use Financial PhraseBank as the primary benchmark and FiQA as external validation. The preparation script writes standardized CSVs into data/processed/.

```
python scripts/00_prepare_phrasebank_fiqa.py
```

If automatic downloads fail, use a local CSV:

```
python scripts/00_prepare_phrasebank_fiqa.py --local_csv data/raw/your_dataset.csv --skip_hf
```

Required schema: text,label. Optional columns: id, source_dataset, agreement, split.

Step 2 - Classical baselines

```
python scripts/10_run_classical_baselines.py --data data/processed/combined_standardized_<timestamp>.csv
```

Purpose: establish majority, Logistic Regression, Linear SVM, SGD, Naive Bayes, Random Forest, and optional XGBoost baselines.

Step 3 - FinBERT baseline

```
python scripts/11_run_finbert_baseline.py \  
  --data data/processed/combined_standardized_<timestamp>.csv \  
  --model_name ProsusAI/finbert \  
  --epochs 3 \  
  --batch_size 8
```

Use GPU for this experiment. For a smoke test, add --max_rows 1000 --epochs 1.

Step 4 - Structured financial signals

```
python scripts/12_run_structured_features_experiment.py --data data/processed/combined_standardized_<timestamp>.csv
```

Purpose: test whether numeric, lexical, negation, uncertainty, direction, and ticker-like signals improve model quality.

Step 5 - Collect external corpora

StockTwits:

```
python scripts/01_collect_stocktwits.py --symbols AAPL MSFT TSLA NVDA --limit_per_symbol 30
```

SEC 10-K filings:

```
export SEC_USER_AGENT="Your Name your_email@example.com"
python scripts/02_collect_sec_10k.py --tickers AAPL MSFT NVDA TSLA --years 3 --max_filings_per_ticker 2
```

Financial news headlines:

```
python scripts/03_collect_financial_news.py --weak_label
```

Optional NewsAPI/Finnhub:

```
export NEWSAPI_KEY="..."
export FINNHUB_API_KEY="..."
python scripts/03_collect_financial_news.py --include_newsapi --include_finnhub --weak_label
```

Step 6 - Retrieval experiment

```
python scripts/13_run_retrieval_experiment.py --data data/processed/combined_standardized_<timestamp>.csv
```

With external corpus:

```
python scripts/13_run_retrieval_experiment.py \
--data data/processed/combined_standardized_<timestamp>.csv \
--external_corpus_csv data/external/financial_news_headlines_<timestamp>.csv
```

Step 7 - Agreement-aware training

```
python scripts/14_run_agreement_aware_experiment.py --data data/processed/phrasebank_standardized_<timestamp>.csv
```

Purpose: use PhraseBank agreement levels as supervision confidence and test ambiguous/low-agreement robustness.

Step 8 - Calibration and abstention

```
python scripts/15_run_calibration_experiment.py \  
  --data data/processed/combined_standardized_<timestamp>.csv \  
  --base_model linear_svm \  
  --method sigmoid
```

Purpose: produce ECE, Brier score, reliability bins, and abstention curves.

Step 9 - CARA-lite integrated pipeline

```
python scripts/16_run_full_cara_lite_experiment.py --data data/processed/combined_standardized_<timestamp>.csv
```

With external corpus:

```
python scripts/16_run_full_cara_lite_experiment.py \  
  --data data/processed/combined_standardized_<timestamp>.csv \  
  --external_corpus_csv data/external/financial_news_headlines_<timestamp>.csv
```

Step 10 - One-command classical pipeline

```
python scripts/90_run_all_classical_pipeline.py --data data/processed/combined_standardized_<timestamp>.csv
```

Step 11 - Share results for comparison

Zip and share the results/ and figures/ folders. Minimum files needed:

```
classical_baseline_summary_*.csv  
structured_features_summary_*.csv  
retrieval_experiment_summary_*.csv  
agreement_aware_summary_*.csv  
calibration_summary_*.csv  
cara_lite_summary_*.csv
```

Scientific warnings

- Do not claim stock-market prediction from PhraseBank alone.
- Treat StockTwits, SEC 10-K, and news weak labels as noisy unless manually validated.
- Do not make SOTA claims until the complete results folder proves them.
- The likely contribution is reliability and decision utility, not only raw accuracy.